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Modeling of Transdisciplinary Engineering Assets Using the Design Platform Approach for Improved Customization Ability

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ABSTRACT

Original equipment suppliers (OES) that develop unique products are continuously faced with changing requirements during both the quotation and product development processes. This challenge is a different reality from companies that develop off-the-shelf products for the end consumer, which use fixed specifications and where product platforms have been a strong enabler for efficient mass customization. However, product platforms cannot adequately support companies working as OES. The reason is that a high level of customization is required which means that interfaces cannot be standardized, the performance is not negotiable, requirements are not initially fixed, and the specific system interacts with, is affected by, or affects other systems that are simultaneously developed in a transdisciplinary environment. The design platform (DP) approach provides a coherent environment for heterogeneous and transdisciplinary design resources to be used in product development by supporting both designing and off-the-shelf solutions. This research describes the introduction, application and further development of the DP approach at an automotive supplier to support the development of customized solutions when traditional modularity or platform scalability do not suffice. A computer tool called Design Platform Manager has been developed to support the creation and visualization of the DP. The support tool has a connection to a product data management database to link the platform model to the various kinds of engineering assets needed or intended to support variant creation. Finally, the support tool was evaluated by the case company representatives showing promising results.

Keywords: Product Development, Engineering Design, Quotation, Supplier, Engineer-to-order, Product Platform

1 Introduction

The engineer-to-order (ETO) industry differs from other sectors in that the customer is often involved in the quotation stages [1]. This situation contrasts with companies developing off-the-shelf products for the end consumer market and where the level of customization (i.e., variant selection) is low. Variant selection businesses often apply a practical approach to identifying and transferring customer needs into fixed specifications that guide product development (PD). The sales and distribution phase is in this case located after the product is developed and produced. However, original equipment suppliers (OES) commonly developing unique products to be integrated into the product of the customer, cannot work in that fashion, since requirements are often directly determined by the client, and hence the sale phase occurs before the products have been developed. It is not uncommon for products to be developed in cooperation with the customer, which is often an original equipment

manufacturer (OEM) or another supplier which is part of a larger supply chain, and for projects to run for several years. During the cooperative development phase, requirements are often added, removed or changed. This has been investigated in the automotive industry by Almfelt, Berglund, Nilsson, and Malmqvist [2] and is said to be a natural process since knowledge is gained and prerequisites change throughout the project. These changes often stem from the complex interplay between the various disciplines and suppliers involved, who use the same interfaces as inputs for their development processes. When designs require changes to the interfaces, other suppliers and thus their designs are affected. The situation consequently requires changes in affected subsystems or changes to the requirements themselves. The increasing complexity of products where mechanics, electronics, and the embedding of code in the products; where products must be developed in cooperation between design, analysis and be evaluated regarding producibility, ultimately puts high demands on companies' abilities to work in a transdisciplinary fashion. As disciplines within companies, such as design, purchase, analysis, aftermarket etc. are centralized to specific departments it becomes more crucial and at the same time more complex to receive an overview of the company assets which have the possibility to be reused in ranges of products and to have a common object to be used for communication.

Product platform strategies, as enablers for customization, have been widely accepted in the industry to serve a wide variety of products while maintaining business efficiency. Early descriptions of product platforms focused on efficiently providing the market with a product variety while also keeping internal variation low and in that way reach a higher level of standardization of the production [3]. Platforms have also served to reach different customer segments efficiently and simultaneously by featuring commonality in product components and interfaces. Recent research has focused on platforms using a more abstract definition; these platforms aim to reuse more of the skills and knowledge (i.e., assets) created in a given company to reach higher efficiency during development. From that perspective, Johannesson [4] has asked whether businesses can afford not to adopt a platform approach. However, most of the existing research on product platforms is focusing on product developing firms with the end consumer as the customer in focus. These approaches tend to require a focused development of the platform and late customer involvement, which in turn requires a forecast regarding which future variants are to be derived from the product platform. There are limitations to developing such product platforms for OES since their business model forces them to develop unique solutions for each new project to satisfy the customer's specific requirements which essentially is their competitive edge. Changing requirements throughout the development process is also a factor that makes rigid product platform definitions hard or impossible to apply for the OES.

This article's goal is to investigate the possibility of applying, supporting and take advantage of a platform approach at an OES where traditional product platform definitions have been shown to be hard or impossible to implement. Therefore, this article, in detail, applies and realizes the design platform (DP) approach, described in [5]. The academic contribution consists of adding to the feasibility and validity of the DP for a type of company that traditionally has been unable to fully take advantage of product platforms. The study also builds upon existing research which has emphasized similar challenges for OES [6] and proposed methods to support their situation [7]. The industrial contribution consists of the application and support of the proposed approach to enable OES to engage in platform development to a greater degree, which has been pointed out as a crucial factor to stay competitive [4].

2 Related work and state of the art

Requirements management is a research field which deals with how requirements are elicited, analyzed and specified [8]. Since customization concerns the fulfilling of individual customer requirements, these two research fields are highly linked. There is a vast body of knowledge in the area of managing requirements and there exist many studies in the areas of consumer products [8] and configuration systems [9]. However, the prerequisites of customizing for an end consumer differ from those for an OEM, and past research has proposed different methods. OES are often required to stretch the boundary with each new PD project, which forces them to explore new ground regarding design and the way they carry out design. The dynamic and transdisciplinary nature of this environment often results in changes to or the addition of new requirements and the elimination of others [2]. Other reasons for changing requirements include incomplete capture, traceability issues, customer-driven changes, incorrect or ambiguous language, missing requirements and redundancy [10].

2.1 Customization in the OES industry

Research has previously focused largely on company and customer integration and collaboration [11], often involving a single business interface [12], which in many cases is an over-simplification of reality. OES are often part of a supply chain where other suppliers and an OEM act in between them and the end customer. This complexity introduces several interfaces and stakeholder interests that the company must manage. Holistic research in this area, taking all or several of these perspectives into consideration, remains scarce. Tuli and Shankar [11] describe lean in collaborative product development and review the existing literature on the supplier, OEM, and customer integration. One interface with the customer that is central to customization is the customer order decoupling point (CODP). The CODP is defined as the point in the flow of goods at which forecast-driven production and customer order-driven production are separated [13]. The CODP is often viewed as a point on a one-dimensional line that can be coupled to the level of customization [14]. Wikner and Rudberg [15], however, propose a two-dimensional categorization for companies in the product realization process, including both the engineering and production dimensions.

To gain efficiency when offering customized products, Vollmar and Gepp [16] study the introduction of standardization in the ETO oriented business. Standardization is however not a possible approach for all companies developing customized products due to the inability to standardize interfaces, that the performance is not negotiable, requirements are not fixed at the outset, and the specific system interacts with, is affected by, or affects other adjacent subsystems that are simultaneously developed by other actors. Design reuse has been coined an enabler for ETO companies to succeed in the process of designing customised products. Brière-Côté, Rivest, and Desrochers [17] propose a support method to incorporate emerging solutions in a generic product structure as a way of increasing design reuse in ETO companies. They used functional, technological and physical levels of abstraction. Baxter [18] considered knowledge to be actionable information and problematized the fact that many previous design knowledge reuse systems exclusively focused on geometrical data, which is often not applicable in the early stages of development. Future reuse models need to contain problem-solving methods, solution generation strategies, design intent and project knowledge. Knowledge-based engineering (KBE) has been pointed out as an enabler for mass customisation to manage large ranges of variant designs as well as respond quickly to customer requirements. KBE, however, needs to be further researched to, for example, develop methodological support, improve transparency and efficiently source and reuse knowledge [19]. Stokes [20] presented a complete framework and a detailed methodology, called MOKA – methodology and software tools oriented to knowledge-based engineering applications, that aim to collect and formalize knowledge to create knowledge-based systems.

2.2 Enabling customization by platform models and methods

A product platform approach can be defined as the development and implementation of technology, components or subsystems that are shared across multiple products [21]. Product platforms have also been shown to prolong the average product life cycle, because of shared components and the flexibility of introducing newer components over time within that same architecture. The authors also state that product variants derived from a product platform, according to this definition, have both higher aggregate sales and aggregate gross profit margins over the product lifecycle compared to products which are not derived from a platform. A more abstract definition is given by Robertson and Ulrich [22]: “The collection of assets [i.e., components, processes, knowledge, people, and relationships] that are shared by a set of products”. Other definitions found in research papers usually fit in between these two. Halman et al. [23] state that companies in the industry have not been keeping pace with research on platforms due to a lack of tools. The authors have also identified a disjointed view of platforms in the industry, which could be at least partly explained by the wide range of definitions that literature has proposed. One risk of using a product platform approach is the tradeoff between commonality and distinctiveness [22]. This trade-off has been subjected to optimization, as summarized by [24]. Another trade-off involves development efforts for the initial platform and the uncertainty regarding whether the right platform has been chosen, such that a sufficient number of derivatives can be developed to recover the added expenses [23]. Managers also need to consider the various trade-offs, which include the degree of heterogeneity of customer needs in the target market, as well as different internal organizational approaches to implementing the platform strategy [21].

A concept which seems to have preceded the product platform and which is highly related is the product architecture [25]. It is defined by Pirmoradi, Wang, and Simpson [26] as “a concept for describing relations among components and connecting the functions to the components in a product.” According to the authors, a platform architecture then becomes “the logical relations between common and unique elements for enabling highly customized products based on customer preference.” According to Mortensen and L  kkegaard [27] product architectures have the following three characteristics: (1) shared core interfaces, (2) core modules/systems exist in balanced performance steps, and (3) architecture(s) are prepared for several future development projects. The authors identify ten central principles for the design of product architectures to cover a particular market. These include isolation of low selling options from the product architecture, decomposition of the product architecture into modules, identification of stable interfaces and more.

Modularization is a critical enabler for mass customization and platform thinking which has emerged from the theories of product architectures. A module can be defined as “a functional building block with specified interfaces, driven by company-specific reasons” [28] and is a central concept in early publications on product architecture. These reasons can include aspects usually connected to the voice of the customer, the voice of engineering or the voice of business [29]. Krause et al. [30] propose a toolkit to use in design as a means to keep the external variety high at the same time keeping the internal variety low to manage design process complexity efficiently. Four principles outline the basis of the proposed toolkit where modularization has an important role: (1) Clear differentiation between standard components and variant components, (2) reduction of the variant components to the carrier of differentiating properties, (3) a one-to-one mapping between differentiating properties and variant components and (4) minimal degree of coupling of variant components to other components. Stjepandi  , Ostrosi, Foug  res, and Kurth [31] outlines several developments and tool implementations of modularity in the context of concurrent engineering. They state that modularity is an essential property of product design as a multidisciplinary concept. They conclude that a current trend is to combine and integrate different technologies such as advanced CAD systems, product configurators, agent-based systems and PDM systems. They identify a need for holistic and concurrent approaches to respond to the challenge of a transdisciplinary environment. Transdisciplinary refers to disciplines crossing boundaries for a dynamic and adaptive system. The solutions produced within a transdisciplinary setting extend beyond traditional boundaries [32] and can be mutually dependent. Thus, a transdisciplinary engineering asset can be considered as a generic asset that can span one or several disciplines in order to be considered and support another discipline.

Although only a small amount of research has been conducted on OES, a platform formulation for a supplier using small batch production has been investigated by Berglund, Bergsj  , H  gman, and Khadke [33]. These authors conclude that a modular platform is not feasible in such a company and that most of the reuse is found at a higher level of abstraction. Platform formulations are investigated among ETO companies in Andr  , Stolt, Elgh, Johansson, & Poorkiany [34] and categorized based on the level of abstraction by which they are defined. Cooper [35] suggests that one deliverable from TD can be a technology platform, which was further investigated by H  gman, Bergsj  , Anemo, and Persson [6]. The authors present a technology platform definition that is not connected to a specific implementation – as a product platform is – but consists instead of design knowledge, product concepts, applied technology, and technological capabilities that support product realization. Gu  laugsson, Ravn, Mortensen, and Sarban [36] describe a tool, which is based on the technology platform concept, called the Conceptual Product Platform. The aim is to communicate the product portfolio by mapping application requirements through concepts onto the product organs, which are the physical features that realize the functions. Levandowski, Jiao, and Johannesson [7] propose a two-stage model of an adaptive product platform as a means to manage changing requirements during the development process. In the first stage, modules are preliminarily configured, and high-level decisions are made. In the second stage, scalable parameters are adapted once requirements are stabilized. Kristianto, Helo, and Jiao [37] propose a system level configurator directed to ETO companies developing large complex products, involving suppliers with development of their own. The authors propose that system level configuration should maintain the key building blocks and possible interfaces between components. This is managed using templates. Each supplier can then provide their building block for the complete design. Platforms and their associated variants pose challenges for many conventional PLM systems on the market. Bruun, Mortensen, Harlou, W  r  sch, and Proschowsky [38] describe a visual architecture representation and its operational handling in a PLM system to enable companies to overcome the challenging situation of identifying common modules when developing product families. A product platform model is a model that describes the generic architecture of the platform and its connection to the derived product variants. A platform model also contains attributes, rules and additional data that is crucial to communicate the platform abilities and limitations of creating product variants. Several product platform models have been proposed in literature which to some extent fulfills the previous

definition. The product variant master (PVM) is a method that can to some extent be used to model a product platform [9]. The main aim of the PVM is to map a company's various product variants and couple them to a generic product architecture to create a foundation for introducing configuration systems. Another methodology that has been termed a platform model, focusing on early stages of design, is the configurable component concept [39], in which the path from functional requirements to design solution is mapped.

2.3 Research gap and summarization

The concept "product platform" seems to have been termed somewhere during the eighties. However, platform thinking [40] including reuse, component standardization and commonality [41] have probably been practiced by design engineers long before the obvious business possibilities was discovered and systematized. Table 1 summarizes some essential aspects of platform thinking from a selection of publications spanning the period from 1985 to 2017. The table takes into consideration what aspect of platform thinking that is targeted and what means that is utilized to do so. Also, what areas that are included in the platform concept is identified as well as what firm perspective the author targets, what case example that is used and what type of company the examples focus on. N/A is used for aspects which were not existing or could not be identified. The authors of [40, 42] identify the business opportunities of platform thinking and include the whole firm in the mindset. These publications, however, lack the detailed support for firms to gain the direct benefits of platform thinking. The response from design departments in later publications focus on modularization and standardization of components and subsystems in products. It is soon discovered that certain types of businesses, like the aerospace supplier companies, have difficulties in applying such approaches. From this perspective, the product platform concept is further developed, and reuse possibilities are sought in other areas than physical components. Today, platforms are used on several levels in companies from early phases of design, as an architect tool, to business tools used for branding in the consumer market. However, the discussion regarding how companies acting as OES, like aerospace and automotive suppliers, can benefit from a platform approach seems to just have been started. There is, therefore, a lack of research that describes how OES can reap the rewards of platform thinking when the rigidity associated with standardized interfaces and pre-planned variants is not an option. Neither the reduction of the number of offered variants, as proposed by Vollmar and Gepp [16], is a possibility for these companies since their competitive edge often lies in the ability to offer unique products. Adding to the challenge of product platform application is changing and unknown future requirements. The concepts of the technology platform [6] and the abstract definition of the product platform [22] are feasible paths for OES to apply platform thinking which would allow them to take advantage of assets from different disciplines that extend beyond physical components. There are, however, not many examples of support methods and tools to aid design engineers in making use of such a platform definition other than using the definition for explanatory purposes.

Table 1. Publications on aspects associated with platform approaches in chronological order

Ref	Aspect of platform thinking	Means	Focus areas in the platform	Perspective	Industry example	Company type
[41]	Commonality	Modular BOM	Product	Business and product strategy	N/A	N/A
[25]	Component sharing and standardization	Functional product architecture using modules	Product	Design	N/A	N/A
[42]	Common elements, especially the core technology	No specific guidance	Several functions company	Business	N/A	N/A
[40]	Core technology, component sharing, similar manufacturing processes etc.	No specific guidance	Several functions company	Business	N/A	N/A
[28]	Commonality	Modules, interfaces.	Product	Management and design	Automotive, power tooling	OEM and OES
[22]	Planning commonality and distinctiveness	Modules, interfaces	Several with a focus on the product	Management and design	N/A	N/A
[23]	Commonality	Customer segments, modules	Product	Business and management	Semiconductors, power tooling, digital printing	OEM and OES
[9]	Product configuration	Modules	Product	Business and management	Cement plant	OEM
[43]	Supplier collaboration and communication	Modules, interfaces	Product	Management	Automotive	OEM

[6]	Reuse, core technologies	Reusability guidance	Manufacturing, materials, designs, methods etc.	Engineering management	Aerospace	OES
[17]	Reuse	Generic product structure	Product	Early stages of design	Aerospace	OES
[44]	Reuse	Function mapping	IT support for an integrated approach	Management	Aerospace	OES
[30]	Commonality	Modules	Product	Design	Multi case	N/A
[36]	Front end platform modeling	Functional mapping to concepts	Novel technologies and concepts	Technology development and design	N/A	N/A
[38]	Product architecture in a PDM system	Modules, interfaces	Product	Design	Multi case	N/A
[7]	Reuse and configuration	Scalable, modular, functional mapping	Product	Systems engineering, design	Aerospace	OES
[37]	System-level configuration	Conceptual	Product	Management and design	Shipbuilding	OEM
[16]	ETO standardization	Modules and framework	Business, technology, process, and people	Business	Plants, power, rail	OEM and OES
[39]	Reuse	Functional mapping, modular, scalable	Abstraction of solutions and requirements	Early stages of design	Aerospace, automotive and power	OES
[21]	Commonality	Customer segments, modules	Product	Business	Lighting	OEM
[5]	Reuse	Generic product structure	Product and design process knowledge	Early stages of design	Automotive, aerospace and special machines.	OEM and OES
[27]	Platform thinking, reuse, commonality	Modular and scalable	Process and product	Design	Several case examples	N/A

3 Research methodology

The research project behind this article includes 4 case companies which have been studied during the period between 2013 to 2016 in which the main research framework used was proposed by Blessing and Chakrabarti [45], a design research methodology (DRM). This framework includes four stages: research clarification, descriptive study I, prescriptive study (PS) and descriptive study II (DSII). From this project, the DP approach was developed and conceptually applied to the companies by completing this framework once as well as iterating once between the last two stages. This is described on an overarching level in [5].

This article focuses on describing the final and detailed DP application in one of the companies included in the previously described project. Elements of action research and systems development [46] have been applied within the DRM framework since the researchers were actively participating in the company as the DP approach and support tool was developed and applied. This article presents the outcome of the final two stages of DRM with an emphasis on the PS phase. PS refers to the stage at which a support to aid in the design process is introduced into the studied situation. In this research, the work has been conducted at two levels; the generic and the case level. The former dealt with the synthesizing of the generic model, while the latter dealt with the application of the generic model at the case company. Prototyping of a software tool was conducted as to formalize the gained knowledge and to synthesize the platform approach. Unstructured interviews were conducted on a frequent basis with design engineers as to collect product and product development knowledge. In this manner different knowledge stemming from different disciplines were identified, traced and formalized. The interviews were recorded and analyzed to model generic product structures and to extract pieces of knowledge relevant to the designing of the products. Engineers within different disciplines than design could also be identified and interviewed for their view and relevance to the design process. DS-II describes the phase at which the support is validated, i.e., tested in the intended environment. The case evaluation provided a final assessment by the case company and was conducted using a self-administered questionnaire. The questionnaire contained questions regarding the impact of the DP approach and support tool on factors like speed and accuracy in the design process, level of design knowledge reuse etc. The evaluation was conducted with engineers within the design discipline as this was the focused discipline to support in this study.

4 The Design Platform approach

The DP information model is composed of different objects related to process, synthesis resources, product constructs, assessment resources, solutions, and projects. The resources are linked to a generic product structure which is shown in the unified modeling language (UML) scheme in Figure 1. The words resource and asset will be used

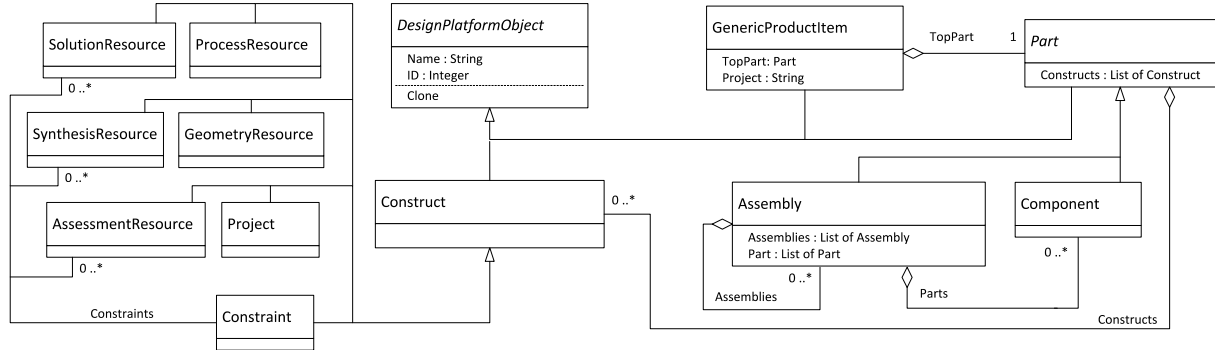


Figure 1. The resources types and relations considered in the DP

interchangeably. The generic representation of the product is referred to as generic product item (GPI) and is represented by the class *GenericProductItem*. Each object that is instantiated from this class becomes a variant and will be unique to a specific project; it consists of its generic structure, cardinality, and attributes. The *GenericProductItem* has the property *TopPart*, which can be one of two types: *Assembly* or *Component*. The *Assembly* and *Component* classes are to be seen as place markers for intended or needed resources to support their realization and not as solutions themselves. The class *Part* has a list of the *Construct* class. The *Construct* is the general type of class that is used to model and points to resources of various kinds. The *Construct* class contains attributes which are summarized in Table 2.

Table 2. Construct attributes

Attribute	Explanation	Type
Description	Description, which is a general explanation of what the Construct is and does	String
Context	Information about when and in what context the construct is valid	String
Technology readiness level (TRL)	TRL denotes the maturity of the construct. This number ranges from 1–9 and originates from the aerospace industry [47]. 1 implies that basic principles have been observed and reported, and a 9 refers to a fully developed and validated construct. The TRL is used to judge which constructs need further development and which constructs can be used as they are.	Integer
Management	Management and responsibility information including relevant people, dates, and versions.	String
Raw data	Information regarding the raw data and their location for the detailed information that makes up the construct.	String

The different types of Construct include *ProcessResource*, *SolutionResource*, *SynthesisResource*, *AssessmentResource*, *GeometryResource*, and *Project* and are to be seen as the building stones of a product variant. These can be differenced differently depending on the needs and possibilities of the specific company where the DP model is applied. The *Constraint* class inherits *Construct*, but it can also be a part of *SolutionResource*, *SynthesisResource*, and *AssessmentResource* describing different limitations and requirements. Objects instantiated from *SolutionResource* relate to finished designs that have formally been designed and have thus been created with some boundaries as defined by *Constraint* objects. *SynthesisResource* has explicit *Constraints* that are related to the method, guideline or tool that the object represents. Objects of the type *AssessmentResource* support the evaluation of product variants and embody mathematical models representing behavior and other properties. They have implicit *Constraints*, the implications of which are made visible through, for example, simulation models. A *ProcessResource* can be in the form of tasks or execution orders of activities that are required or intended to support some part of the design process. Objects of the type *GeometryResource* are commonly parametric CAD models that can span a design space. The resource types are associated but not limited to different disciplines seen in Table 3.

Table 3. Resource types and disciplines

Resource type	Examples of disciplines
ProcessResource	Project management, quality engineering, and general management
SolutionResource	Design engineering and purchasing
SynthesisResource	Design engineering and technology/concept engineering
AssessmentResource	Calculation-, manufacturing-, sustainability- and cost engineering
GeometryResource	Design engineering
Project	Project management, cost-, and quality engineering
Constraint	Requirements engineering and manufacturing

The platform utilization is divided into two stages: expansion and use. The first refers to when the model is established or modified. This stage is supported by a set of activities needed to define a DP as seen in Figure 2. The first step is focusing on organizing the management of the DP and refers to pointing out areas of responsibilities and associated personnel along with setting up a structured database environment. The next step is to formalize

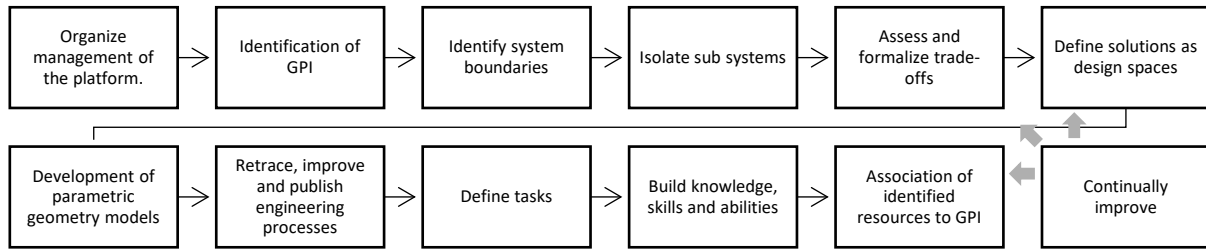


Figure 2. Activities that support the DP definition and expansion

and generalize the product concept by identifying the GPIs. These are the common bases to which different constructs are to be linked. These items might not correspond to a pre-existing design; however, they must be included in the finished design. Next up is to identify the boundaries of the GPIs in terms of what interfaces that interact with the customer's product and which do not. The next step then becomes to isolate the subsystems that only interfaces the own product. These possess a higher level of possibility to be standardized into modules which are reused in several product variants. As these modules are kept isolated they can be further developed in technology development projects which are not directly customer order-focused. Similarly, the subsystems which interface the customer product are identified. These will most likely need to be designed for each new customer order and are the ones in focus for the continuation of the steps defining the DP. Generalised trade-offs associated with the GPIs needs to be assessed as the next step in the process. These concern properties in existing designs and concepts that are related to each other (e.g. weight vs tensile strength) that are known from previous experience or simulations. Continuous or discrete design spaces are defined or identified in relation to the different items in the GPIs. By identifying and modeling the feasible parameter space of a design, variations of such designs can be reused and cover a larger application area. To support the modeling of these design spaces, geometric models are developed by identifying generic characteristics of existing geometry and trends in geometry adaptation. Processes, best practice methods, guidelines and tools for their completion are retraced, modeled and published. These are described using standard classes. Competence teams are established to build and preserve knowledge, skills, and abilities. The GPIs are then linked to the identified descriptions, models and other engineering assets needed for their realization. Improvements through the experience and knowledge gained from PD is continually assessed and applied when a sufficient level of maturity is reached.

The use phase refers to when the model is instantiated, and variants are created. The main steps are illustrated in Figure 3. Early in a project, the DP can be used to find generic process support to guide the design process. Furthermore, similar variants currently or previously in production are determined or identified by searching among

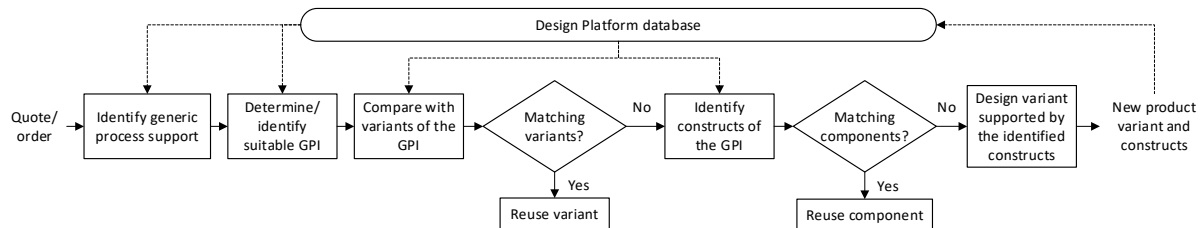


Figure 3 The main steps of the DP use phase

existing GPIs. If no variants exist that correspond to customer requirement, the constructs of the GPI are investigated. If no components can be reused, the quoted product must be designed supported by available constructs. Since resources from different disciplines are linked to the GPIs, these can be located and reused in the design process. These resources include e.g. guidelines, tools, constraints, and more that supports the design of the item.

5 Applying the DP approach

The DP approach [5] has been introduced and tested at a company acting as a 2nd tier supplier in the automotive industry as to investigate its feasibility. The company employs 600 people at the studied site and 10,000 people worldwide. The focus was on a business area in which the company develops and manufactures uniquely customized car interior subsystems. One product concept developed within this field of activity is a pneumatic seat support system, the main aim of which is to vary the pressure distribution between the body and the seat to increase comfort and ergonomics. The core technology includes inflatable air cells which are used to create the varied pressure distribution. The other main part types of the system consist of carrier plate, valves, a pump, and tubes. Figure 4 shows a seat structure on which a pneumatic massage system has been mounted. A pump unit creates a pressurized air flow and is connected to the pneumatic on/off valves via the tubes, which also connect the valves to the air cells. There is usually also a printed circuit board (PCB) attached to the valves, sensors, and pump to regulate the opening and closing sequences of the valves and the pump speed. The valves and tubes are connected to the carrier plate that is mounted to the seat structure.



Figure 4. A seat structure with on which the product in focus has been mounted

The company is in principle willing to customize or redesign any part of the system if it is required to not miss out on business. This inhibits them from standardizing components and modules to a high degree. The level of customisation which is offered is high and therefore speed and accuracy in the quotation process is a key issue. Speed is paramount for returning an offer to the customer quickly, while accuracy is crucial to setting a price that will balance the client's budget realities and the case company's profitability standards. It is therefore vital to reuse as much in the way of pre-existing development engineering assets as possible, which places a heavy demand on finding the necessary information and judging its applicability accurately.

The company has a mapping between projects and components using a traditional file structure in its PDM system. There have also been attempts to describe some of the expertise regarding the process and product knowledge using standard templates. These are, however, unstructured and are not used to a significant degree. A high-level design process description has been defined by the company; however, it is not detailed enough to support the designers.

5.1 A platform support tool and PDM

To support and evaluate the expansion and use of the DP approach, the Design Platform Manager (DPM) support tool was developed. DPM consist of functionality to model GPIs and instantiation of variants and to create associations with the engineering assets used for their realization which are hosted by a PDM system. Figure 5 shows the principle system architecture, core functions, and how the DPM and PDM coexist. The DPM is not intended to take the place of a PDM system but to add to its functionality by being a platform modeling extension to the way a PDM system is commonly used. For this reason, the DPM and the PDM system communicate with the same database. This approach also serves to guarantee that data redundancy is minimized and to enable multi-client and concurrent usage. The use of the DPM does not require a change in the methodology or structure employed in the

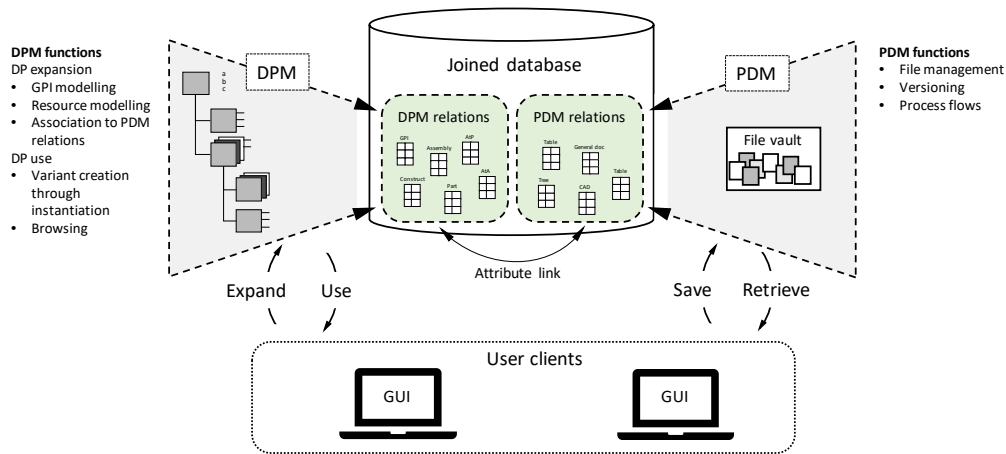


Figure 5. The principal architecture and function of DPM and the PDM system

PDM system. The PDM system focuses on file management, keeping track of versions, metadata, and process flows. To execute the DPM, certain changes *are* required to the PDM database: the addition of database tables that makes it possible to store the different object types included in the DP. This involves the addition of tables for the GPIs and its variants, assemblies, and components. To associate an object created via DPM with objects belonging to the PDM system, a string attribute link was used to point to the specific database record and table in the PDM database. Additional database tables were created to allow for many-to-many associations to be created between objects. Design elements can be used in several GPIs and vice versa which requires specific database tables to specify the association.

The software tool is based on .NET and uses many of the standard forms to display trees and lists. The DPM enables the user to model GPIs as trees whose main structures are composed of assemblies and components as holders of engineering assets. Each level in the structure can have associated assets that are visible either as nodes in the tree or as items in the Engineering Assets view. Different GPIs can be viewed, added, and altered. The variants belonging to a specific GPI can be listed and searched through the Design Instances view. The properties of objects in the user interface can be viewed and changed. To create a functional prototype, a PDM viewer was integrated to enable easy access to the PDM content. This was possible by reading from a PDM database table containing all base classes which also pointed to the database table containing all instances of each class. The database tables could then be searched by sending simple string values to the database. The objects displayed in the PDM database viewer could then be associated with the GPI and variant structures. Further, GPIs, variants, assemblies, and components were all saved, retrieved and updated using specific DPM tables which were created in the PDM database. Figure 6 shows an instance of the application user interface where the GPI “4 way” has been selected. Consequently, the engineering asset objects associated with that GPI level is displayed along with some lower structural levels. The work of identifying GPIs and engineering assets will be outlined in the following.

5.2 DP expansion

The expansion step of the DP was supported by the activities outlined in Figure 2. As a result, a group consisting of two researchers and three company representatives was created to oversee the DP and the work associated with creating and using it. The next step included modeling of GPIs which was enabled using the PVM approach [9]. By interviewing designers and examining several products that had been developed in the past, several generic types of structure and component types were identified. Attributes used to describe and differentiate these components and structures were formalized. The types could be mapped to varying levels of functionality since they would be offered to the customer from basic (static support) to high-end levels (dynamic support of the lumbar, sides, and thigh, along with spine massage). Figure 7 shows a selection of existing variants with different functionalities. The variants that stemmed from the identified GPIs were considered to be instance objects and were linked to the GPI classes. The identified GPIs were saved and structured manually at the outset. The identified GPIs are outlined in Table 4.

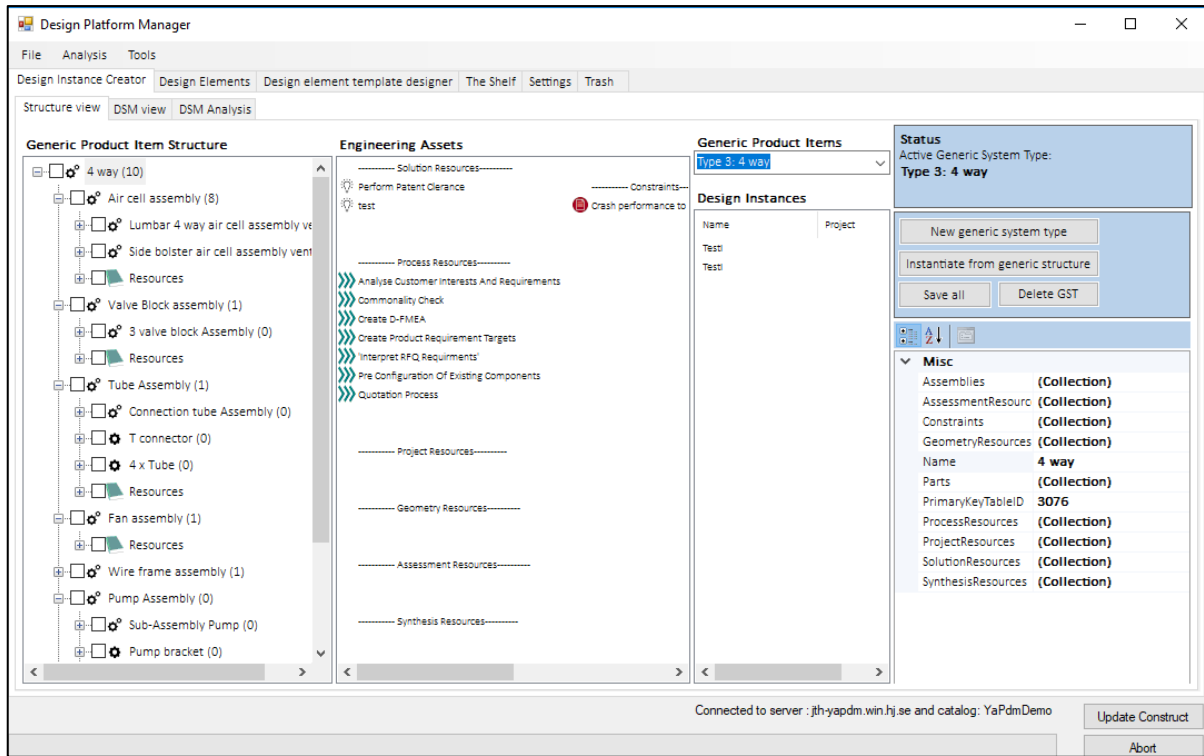


Figure 6 A screen shot of the application Design Platform Manager

The following step regarded the identification of system boundaries. This was performed by investigating a set of product variants and focusing on how the product was interfacing the seat structure. Mechanical interfaces such as attachments were analyzed as well as electrical interfaces such as couplings. This activity directly leads to the next one which deals with identifying and isolating the subsystems which are interfacing the customer's product and thus are customized for each project. Similarly, subsystems which only interface the own products could be identified. The chosen tool for performing this activity was a set of design structure matrices (DSM) and domain mapping matrices as described in [48]. The various parts of the GPIs were associated with input parameters and design variables. For explanatory reasons, Figure 8 shows an example of three principle DSMs and the mapping matrices where the lower left matrix included input parameters that stem from requirements or other types of parameters that are input to the process and thus cannot be affected by the designer.

Table 4 Description of GPI types

GPI types	Description
Static support	This type of GPI cannot be dynamically controlled by the user; it contains only a back plate.
Two-way	This is the simplest dynamic GPI; it enables control only over lumbar back air cell inflation and deflation.
Four-way	This GPI allows a higher level of control over lumbar back air cell support; the components carrying the main functionality include one to three air cells, depending on customer requirements, pump, valves, PCB, tubes, and other supporting components.
Four-way and side bolster	This GPI includes the functionalities of the four-way, but also introduces air cells into the sides of the seat at the same height as the lumbar. The total number of air cells ranges between three and five.
Four-way, side bolster, and ventilation	This GPI adds the functionality of ventilation in the seat. The number of components is the same as above; the difference lies in the design of the lumbar back air cells since air must pass through. The details of the ventilation system are managed by another design team.
Four-way, side bolster, ventilation and massage	This GPI adds a massage functionality that applies all along the spine. The massage part can contain 10 or more additional air cells, with two valves per cell required to manage inflation and deflation. A larger pump is needed to manage the increased airflow and pressure.

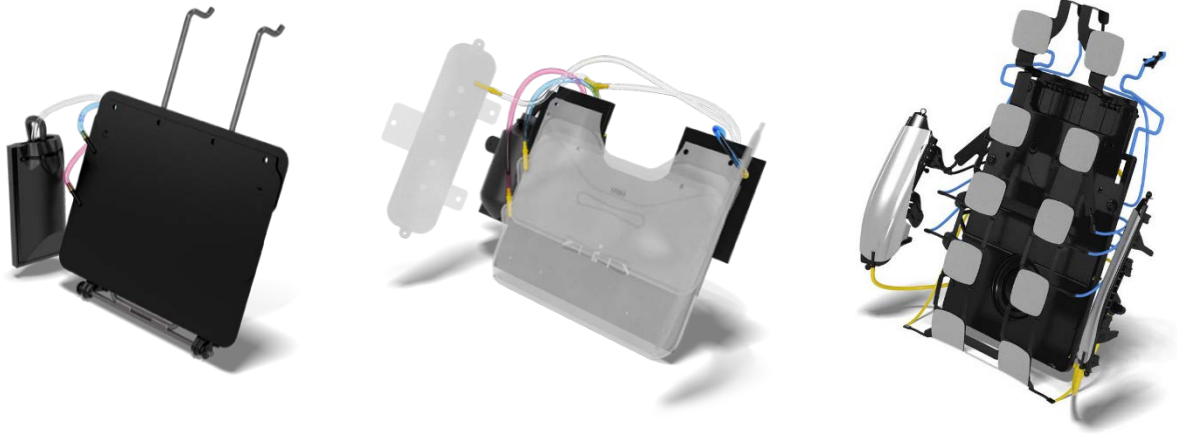


Figure 7. An assortment of variants of the pneumatic seat support system, ranging from simple to high-end functionality

These can concern durability, temperature, weight, sound etc. The middle DSM show the items which are part of a GPI which can be represented by air cells, pump, hose etc. The top right shows the design variables that are set by the designer to design a component or subsystem. Examples of these are air cell dimensions and shape, material thickness, type of material, hose routing etc. The DSMs show the connections in between a domain while a mapping matrix shows the relations in between two DSM domains. The result of the activity showed that it was mainly air cells and carrier plate that were customized for each new variant as well as the inclusion or exclusion of different subsystems to reach various levels of functionality. It was thus found that the various pumps, valves, and tubes have a higher possibility to be standardized. These can be managed as modules and further developed and optimized outside of customer targeted projects.

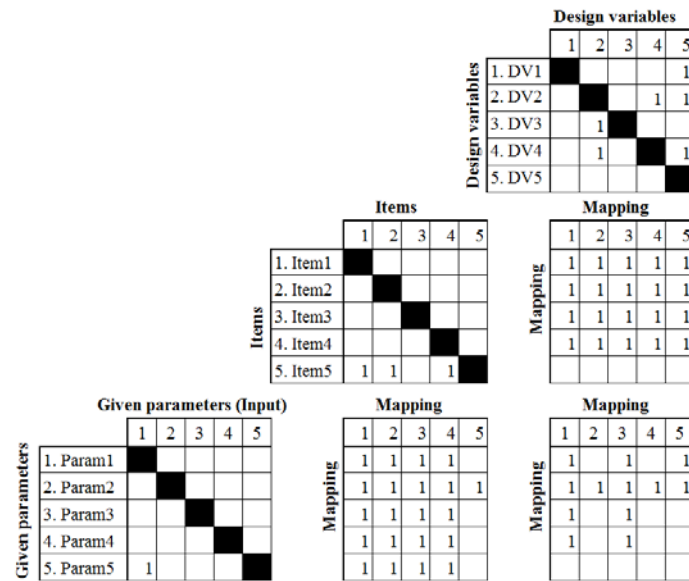


Figure 8. Examples of DSMs and mapping matrices

The additional generic activities described in Figure 2 were applied in an iterative manner as knowledge regarding the GPIs and variant was revised. As to get an understanding of the company process under consideration, the quotation process was mapped through a set of interviews with designers and project managers. The mapping was carried out from a designer's perspective, investigating general activities concerned with the quotation and product design stage. The interviews also revealed knowledge to be considered which stemmed from other disciplines than design. From using the same DSM methodology as described previously, possible clusters to be encapsulated in more detailed activities could be identified. This process is further described in André et al. [49]. Figure 9 summarizes the main quotation process stages that were mapped. It was evident that the quotation stage could be divided into two parts, the first of which concerned an initial configuration using existing components to achieve

a product that was functional enough to demonstrate to the customer as a baseline prototype. If the customer decided to continue, the solutions would be refined and new designs would then be created to meet all customer requirements. A quotation and PD project can take as long as approximately two years, during which time OEM requirements change continually.

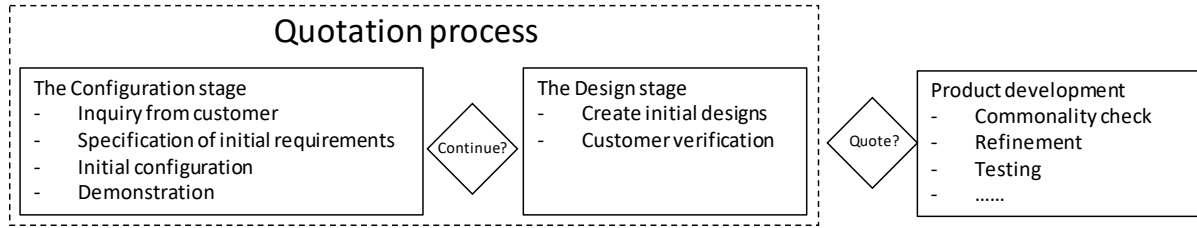


Figure 9. High-level quotation process and subsequent product development

To support the additional steps in Figure 2 and to create a format to describe trade-off curves, engineering processes tasks and more, design elements were introduced in the company as a way of formalizing and expanding the utilization of asset types from several disciplines. The design elements were inspired by the MOKA methodology [20] and consisted of four types: *activity*, *entity*, *constraint*, and *rule*. They can be seen as modules in which design knowledge from different disciplines can be encapsulated, modeled and taken into account during early design stages. The set of attributes distinguishes design elements from MOKA, as does the fact that the relations between design elements are not specified within the design elements themselves. Design elements are standalone objects that can be reused in and applied to different items. Design elements enable the definition of design spaces at a higher level of abstraction than geometry and can be described at various levels of maturity. The types of design elements both share a common set of attributes and unique individual attributes that differentiate them from one another. The common attributes are inherited from the Construct class (Figure 10). The *entity* is a description of an existing embodied component or subsystem; it includes attributes such as function, behavior, and links to geometry models. *Activity* is used to describe a task or process that often includes an execution order; attributes include inputs, outputs, triggers, and objectives. *Rule* outlines a guideline or a set of logical relations for the designer to employ; rules can be described by mathematical formulas, tables, or in text form, and rules can describe design parameters and how they affect different variables. *Constraints* can be one of two types; internal constraints describe limitations that are usually based on some boundary condition, such as manufacturing equipment, while examples of external constraints are customer requirements and legal requirements.

The interviews revealed a set of design elements that could be formulated. They were modeled using template spreadsheets containing attributes and outlining the descriptions using text and images. Table 5 shows a selection

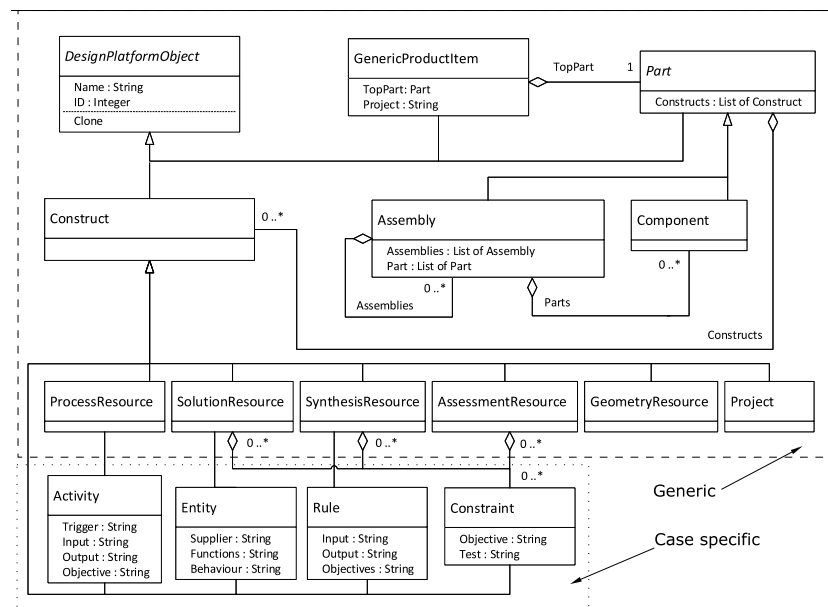


Figure 10. The UML scheme of the generic and case specific additions to the DP model.

of the identified design elements and from which disciplines the knowledge stemmed from. The diversity of disciplines connected with the knowledge of each design element, makes each design element a transdisciplinary engineering asset.

Table 5. Selection of identified design elements

Design element	Name	Disciplines	Description
Activity	Quotation process	Design-, manufacturing-, cost engineering, sales, project management	This design element outlines the mapped quotation process in detail, including the associated tasks in need of completion.
	System requirement breakdown	Design-, requirements-, systems engineering	This design element describes the process that occurs when the key person responsible for the design works with the team to craft a breakdown of all requirements and associates them with the different system levels of the desired product.
	Pre-configuration of existing components	Design-, manufacturing engineering	Existing pumps and valves are reviewed for suitability. If possible, they are also modeled in the seat in a CAD environment to investigate the physical space possibilities.
	Create D-FMEA	Design-, quality engineering,	A system-level design failure modes and effects analysis (D-FMEA) is created based on customer requirements defining functions and compliance with functions. Failure modes and the like are added in accordance with a pre-existing standard; at the component level, existing D-FMEAs will be used as baselines.
	Create air cell CAD model	Design-, manufacturing engineering	This design element describes the process by which an air cell CAD model should be created.
	Perform patent clearance	Design engineering, patenting	This process involves the designer's undertaking an initial investigation to see whether a new solution infringes on existing patents; the design is then passed to the legal department to investigate whether it can be patented.
Constraint Internal External	Noise level	Design-, requirements-, calculation engineering, sales	This constraint is usually part of the customer requirements; it establishes the maximum sound level that the system can emit.
	Temperature range	Design-, requirements-, calculation engineering, sales	This constraint is usually part of the customer requirements; it establishes the acceptable ambient temperature range.
	Lead-free solder	Design engineering, general management	An environmental policy at the company bans the use of lead in solder.
	Machine limitation	Design-, manufacturing engineering	Different machine limitations apply to air cell welding.
Rule	Bolster air cell design guidelines	Design-, manufacturing engineering	The guidelines present shapes, material thicknesses, material choices, and air cell combinations that all serve to bolster air cells.
	Lumbar air cell restrictor design	Design-, manufacturing engineering	To achieve the requirement that air cells can take different shapes when inflated, several types of restrictors can be welded to the air cells. This design element describes the different kinds and offers general guidelines.
	Air cell tube design	Design-, manufacturing engineering	Certain limitations regarding tube bend radius and attachment to air cell apply; this design element outlines these restrictions.
Entity	Pump and valve	Design engineering, purchasing, supply chain	These components are seldom customized. The variants from which to choose have either been developed in TD projects or purchased from a supplier. The requirements regarding air flow and pressure drive the decision-making regarding pump type. Design elements were created for each existing variant and sub-assembly containing supporting structures, such as fir tree clips and cable ties.
	Air cell	Design engineering	These components are always customized for each customer, due to differences in seat geometry. Design elements were created at the single air cell assembly level.
	Tube	Design engineering, purchasing, supply chain	Design elements have been set up for the tubes and the different connectors that were needed. A few variations in diameter exist; the main difference between tube variants is the length.
	Carrier plate	Design engineering	The carrier plate variants were used to create design elements. The number of carrier plates is determined by the number of systems developed for a single car model. A simple variant is normally used for static support. Higher stiffness is needed for the two-way and four-way systems, with the massage systems requiring the stiffest carrier plate.
	Supporting components	Design engineering, purchasing, supply chain	Several design elements were also created for supporting structures intended to hold the system together and prevent noise and dirt contamination.

This section described the expansion phase of the DP approach. Approximately 100 design elements were identified and formalized on spreadsheets. Since the generic nature of some of the design elements, they could be reused on several GPIs. Therefore, the number of relations between assemblies/components and design elements were

many more. Figure 10 further describes the addition of design elements to the generic information model presented in Figure 1. The *ProcessResource* where realized through the modeling of the design element *activity*; *Entity* design elements formalized and expanded the metadata of *SolutionsResources*; *Rule* design elements formalized *SynthesisResources*, and *constraints* were used to model different limitations regarding the previous resource types.

5.3 Intended use of the DP model

To access the design elements in DPM, the spreadsheets on which they were created was saved using the PDM system. The PDM system, therefore, administers the files; keeps track of versions and the file structure. The design element objects could then be found using the PDM viewer in DPM and associated with the GPIs. Other identified generic resources in the PDM system were similarly associated with the identified GPIs and variants. This process also included resources that were used very rarely, since only a few people at the case company knew about their existence and location; making them known and available for other designers.

The following is an example of the intended DP use-phase and the application of the steps described in Figure 3 in the case company design process. When a quotation arrives (Figure 7), requirements are often vague and presented in subjective terms. The design element representing the quotation process is then selected by using DPM for guidance through the different steps. It is then relevant for the design team to get an overview of what variants that have been created in the past which also have similar functionalities and have fulfilled similar requirements. At the beginning of the quotation process, a pre-configuration is made using existing components. DPM is employed to obtain an initial view of what exists and what does not regarding designs, tools, and methods. Components that are seldom customized, such as pump and valves, are picked from existing variants that offers the desired functionalities. Supporting components are also picked from items currently in production. To complete the pre-configuration, several air cells are needed to produce the level of back support intended. These are chosen based on shape, with the most suitable selected for a rough prototype by searching among the *entities* that describe air cells. This step concludes by demonstrating the prototype for the OEM, which in turn evaluates the system, based on its requirements and reactions. Using the OEM's input, specifications are more richly defined, and the development team enters the design stage of the quotation process, at which point refined solutions are required. Suitable GPIs are identified for browsing to find resources that support the fulfillment of customer requirements. Each structural level might contain resources for the realization of a specific item. Variants corresponding to the GPI can also be searched to determine whether similarities exist. It is likely that pumps and valves used in the pre-configuration can be employed, even in the refined solution. For some levels in the GPIs, there might be associations with finished designs; this is especially likely for supporting components and tubes. For other structural levels, there might be no associations with finished designs; specifically, this is common with components and systems that interface with the customer's products and when subjective requirements cannot be quantified. It is unlikely that entities can be found that already complies with the required seat geometry. The items in the GPI corresponding to the lumbar, bolster, and massage air cells are reviewed. Rules describing guidelines are found regarding the outer shape and air cell restrictor designs. Constraints regarding the minimum welding radius also apply and must be considered when designing the new air cells. An activity design element relating to the creation of an air cell CAD model is also available. The identified resources are selected using DPM and instantiated, which creates a new variant. A new design is demonstrated to the OEM, and the decision whether to quote is made. If a contract is signed, the DPM variant model is brought into the PD process. All along the development process, requirements tend to change regarding performance and interface geometry. At these times, the DPM is used to browse the relevant items and find resources that enable responding to the change. There might be tools to assess the height of inflated air cells or tube routing rules. Both the resources and designs populate the model as new knowledge and designs emerged throughout the project. The knowledge which is gained expands the DP by using lessons learned to develop knowledge from issues that arise during projects and creating new design elements. Also, existing design elements and other resources should continually be improved by using the knowledge gained to create new revisions and by using other suitable formats that are coupled to the GPI and the variants.

6 Discussion

In an era of increasing individualization and product complexity; greater demands are placed on OES. They must excel in managing the dynamic customer requirements and have a way to adapt to changes, at the same time as

being accurate and quick during quotation and design processes. This takes place in a transdisciplinary environment where company functions are centralized which can lead to severe implications when requirements change.

This article has contributed with the detailed application of the DP approach [5] in an OES. The focus of the study has been on suitable formats to embody the different asset types and how these can populate the DP to make them available and to increase the level of reuse and thus the efficiency of the customization ability. The DP, similar to the technology platform [6] or platform thinking [40], focuses on making use of the abstract definition of product platform given by Robertson and Ulrich [22]. Recent research has, however, left a gap when it comes to supporting such platform definition and making it useful when customizing products. Early publications on utilizing product platforms have had a business focus and included whole firms [42]. They have however lacked detail when it comes to describing how to apply these ideas in design and manufacturing processes. Later publications have had a detailed focus on the artifact as a way to gain the business advantages of platform thinking [30]. There are examples of approaches utilizing a higher-level platform definition, e.g., where modules are both scaled and configured conceptually in Levandowski et al. [7] and in [39]; however, detailed examples and case applications are few to find. The DP introduces and allows for different knowledge formats, ranging from physical components to more abstract descriptions, to be included in the platform; making it useful for customization in several ways. The use of the DP is intended for but not limited to companies that are developing highly customized products. Even though a company has several functions, all serving the manufacturing and delivery of products, the GPI is a suitable view for the DP to be based upon, as in the PVM approach [9], since the product is “the heart of the company” according to [17]. The GPI remains the common denominator between different company functions which makes it a suitable carrier of knowledge from an array of disciplines. Knowledge from disciplines like manufacturing-, calculation-, quality engineering and purchase can be described and coupled to the generic product concept for early and concurrent consideration and support during quotation and design. The DP also differentiates between what is generic and instance specific. This is opposed to only use a project structure which requires knowledge of previous projects from users to find applicable information.

The DP model is owned by the design department, managed as an asset in of itself; continually updated and maintain to guarantee its usefulness. The complete DP approach, however, is an overarching approach for the whole company and a way to systematically view and work with the enterprise resources; making them available and increasing the possibility of knowledge reuse.

The expansion phase is about being proactive during PD by finding best practices and being zealous when creating the descriptions to populate the DP. It is at this stage that the platform evolves by adding new knowledge that can serve as resources for future projects. By introducing generic resources in formats such as parametric CAD models, task descriptions, and tradeoff curves enable a set-based approach that defines spaces rather than point-based solutions. Using the DP aids development loopbacks when iterations must be undertaken and to a higher degree supports knowledge reuse and design process standardization. Once a design has been created, tested by the customer, and judged not to comply with one or more requirements, the DPM can be used once again to search for resources that might solve the remaining issues. Where a component could theoretically be reused, the new situation might demand an increase in performance that the existing component cannot meet. In this event, the GPIs can again be searched for resources to be utilized in the design of the desired component or for the modification of a specific attribute. Due to the connection between variants and GPIs and the associations with resources, the DP evolves as new information is added to it.

To receive the opinion of potential users, the DPM was used on company-specific data. The DPM was then demonstrated to the potential users which consisted of an engineering manager, a project manager, a lead engineer and a designer. They were then allowed to comment and grade the performance of the support tool and associated working approach using individual questionnaires. The overall judgment was positive. Among the company representatives' anticipated effects of implementing DPM was a decrease in project time due better overview and access to the different resources. The need for formal design loops was also believed to be reduced. The company representatives emphasized that DPM would increase the level of support for the designer as well as increasing the level of knowledge reuse. The areas pointed out as needing additional work was focused on the user interface. The manual input to the system was seen as time-consuming; as was the need for a structured working approach to be employed across the company on a global scale. A more detailed evaluation of the DP approach which includes the presented case company can be found in [5].

7 Conclusions and future work

This article has targeted an area in platform-based development which has been a blank spot in academia and industry. OES are challenged trying to apply traditional product platform models and methods due to unknown and changing future requirements. Based on this gap, this article presents the application of the design platform approach, consisting of design knowledge in combination with complete solutions. This platform provides a coherent environment that can be used to develop, manage, and use corporate assets systematically in the OES context. The model can describe both the company's current state, and its future target condition while also pointing to difficulties and limitations. A focus was also placed on the integration and practical approach associated with a support tool and PDM system.

The DP approach is an enabler to gain the benefits of platform thinking both considering the engineering assets residing in a company and as a formalized model that can be supported by IT applications. The DP model provides construct types that can be categorized and help identify and describe company assets. The associated activities support the definition and expansion of the DP. The GPI is a suitable way to model a product concept and to be used as a common view and carrier of knowledge. In turn, design elements are suitable to be used as carriers of product and process knowledge and to be associated with the GPIs and specific variants. The DP approach and support tool have undergone an evaluation that showed a promising overall result regarding decreasing the time spent on projects, the need for formal design loops, the level of support for the designer, and the level of knowledge reuse. The evaluation also pointed to areas that could be improved, such as visualization and the level of automation. It can be concluded that the DP approach is a feasible path to increase the reuse to support customization for the studied EOS. Similarly, the generalized application of the DP to similar companies is discussed to be a suitable approach.

Future work will include further development and refinement of the DP model on a general level. The model will be applied to more companies to test its generalizability further. The focus will be on modeling the process to support the use phase including modeling of relationships between assets to not only couple them to items but also execution order. The DPM support application will be further refined regarding functionality and usability, as suggested by the case company representatives. The support tool will also be introduced at other companies.

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